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EXPERIMENTS ON PACKET CAPTURE TOOL – WIRESHARK

AIM:

To study the features of Wireshark as a packet capture tool and understand the encapsulation of information at various layers of the protocol stack.

ALGORITHM:

1. Download and install Wireshark on the computer.
2. Open Wireshark and observe the available network interfaces.
3. Select the active network interface through which data is transmitted and received.
4. Start capturing network packets by double-clicking the selected interface.
5. Generate network traffic by opening a website or executing the ping command.
6. Observe the captured packets in the Packet List pane.
7. Select a packet to view its protocol details in the Packet Details pane.
8. Examine the packet bytes in the Packet Bytes pane.
9. Apply display filters such as `dns`, `tcp`, `udp`, and `icmp` to identify specific types of packets.
10. Observe the encapsulation of data across the Ethernet, IP, transport, and application layers.
11. Stop the packet capture and save the captured data for further analysis.

Commands used to generate network traffic:

```
ping google.com
```

Display filters used in Wireshark:

`icmp` – Displays ICMP packets.

`tcp` – Displays TCP packets.

`udp` – Displays UDP packets.

`dns` – Displays DNS packets.

`http` – Displays HTTP packets.

OUTPUT:

Wireshark captured network packets transmitted and received through the selected network interface. The captured packets displayed source and destination addresses, protocol types, packet lengths, and other protocol details.

The packet details showed the different layers involved in communication, including Ethernet, IP, TCP/UDP, and application-layer protocols, depending on the selected packet.

RESULT:

Thus, Wireshark was used to capture and analyse network packets successfully. The features of the packet capture tool were studied, and the encapsulation of information at various layers of the protocol stack was understood.